

Business Intelligence at Mitracomm



How to ?

Optimalisasi Layanan Mitracomm

Core Bussiness

Internal : Business Process, Transactional Process,
Value Added Services

Data

External : CS Agent, Omnichannel,

Data (SOP Transaksi, User, Keluhan
User), Format Data

System Existing

?

Problem

Profit, Network, Structure, Process, Product System, Product
Performance, Product System, Services, Channel, Brand, Customer
Engagement

Solution

?



Services Company

Business Process Services

- Contact Center Outsourcing
 - Full Managed Services
 - Partial Managed Services
 - Labour Supply
- Digital Engagement

IT Solutions

- Hosted Contact Center
- Omnichannel
- IT Managed Services

Transactional Process

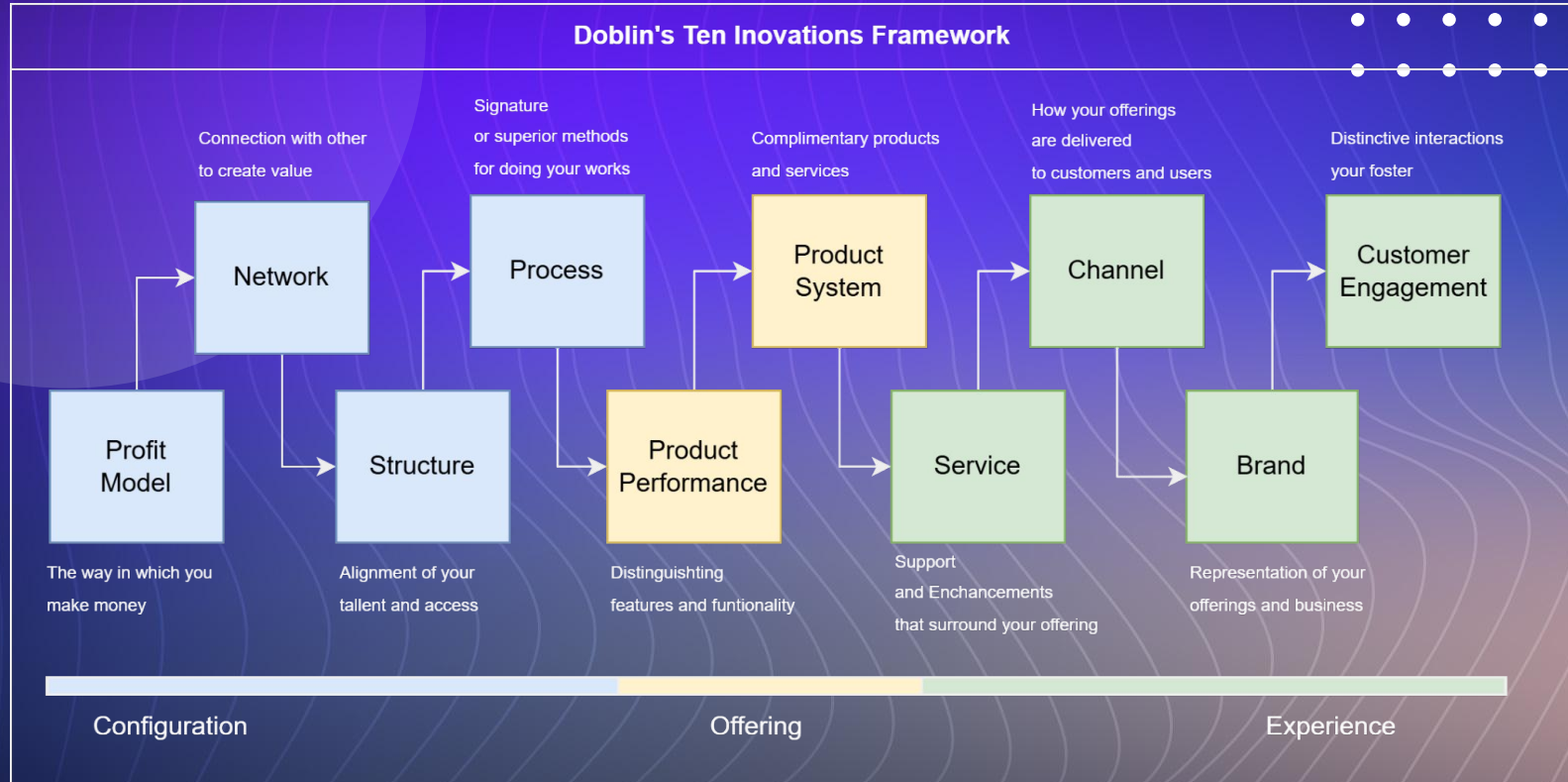
- Walk in Customer Services
- Assessment Process
- Data Entry
- Loyalty Managements
- Direct Sales
- Outbound call Services (Telecollection, Telemarketing, Telesales, Welcome Call, Data Cleansing, Telesurvey)

Value Added Services (VAS)

- Training (Motivational & Softskill)
- E-Learning & Premises
- Learning Management System
- Asset Management System
- Employee Management System
- Laptop Rental



Doblin - 10 Types of Innovation



From Larry Kelley, one of the Doblin co-founders

01. Data Management ?

Warehousing

Repository terpusat yang menyimpan data historis dan terintegrasi dari berbagai sumber untuk mendukung analisis dan pengambilan keputusan.

Tujuan :

Repository terpusat yang menyimpan data historis dan terintegrasi dari berbagai sumber untuk mendukung analisis dan pengambilan keputusan.

- Integrasi data dari sumber heterogen.
- Mendukung query dan analisis kompleks.
- Memisahkan proses OLTP (Operasional) dan OLAP (Analitis)

Characteristics

01

Problems

Data dikelompokkan berdasarkan subjek (e.g., penjualan, pelanggan).

02

Integrated

Data konsisten dari berbagai sumber.

03

Non-Volatile

Data tidak diubah, hanya ditambahkan / dibaca.

04

Time Variant

Data historis tersimpan dalam timestamp.

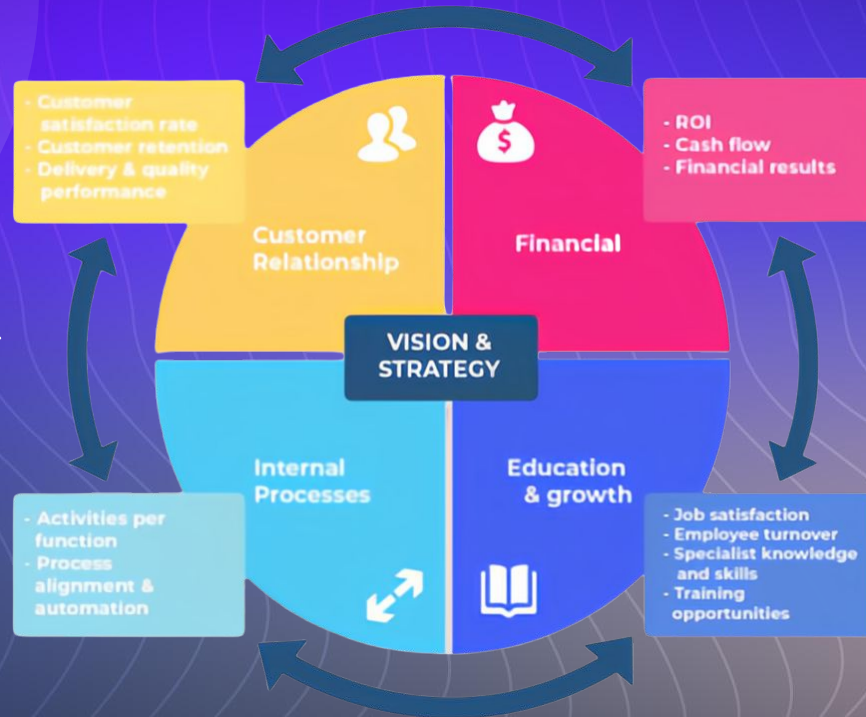


Balance Score Card

Assessing
Organizational
Success.



Existing ?



Metabase/
Superset

Visualization



Action

Allert
Services
ext



Right decision = Organizational Success



● GOALS AND OBJECTIVES

Tujuan jangka panjang dan target spesifik yang harus dicapai

● KEY PERFORMANCE INDEX (KPI)

Metric untuk mengukur tercapainya tujuan dan target

● PROCESS AND PROCEDURE

Proses dan prosedur yang dilakukan untuk mencapai tujuan dan target

● RESOURCES AND CAPABILITIES

Proses dan prosedur yang dilakukan untuk mencapai tujuan dan target

● RISK AND MITIGATION STRATEGIES

Proses dan prosedur yang dilakukan untuk mencapai tujuan dan target

● PERFORMANCE METRIC AND MONITORING

Proses dan prosedur yang dilakukan untuk mencapai tujuan dan target

● CONTINUOUS IMPROVEMENTS

Proses dan prosedur yang dilakukan untuk mencapai tujuan dan target



Memahami kebutuhan bisnis
Desain struktur data yang tepat
Penggunaan ETL
Pengelolaan data yang teratur

CRISP-DM Method

Cross Industry Standard Process For Data Mining

RIGHT DECISION = RIGHT ANSWER

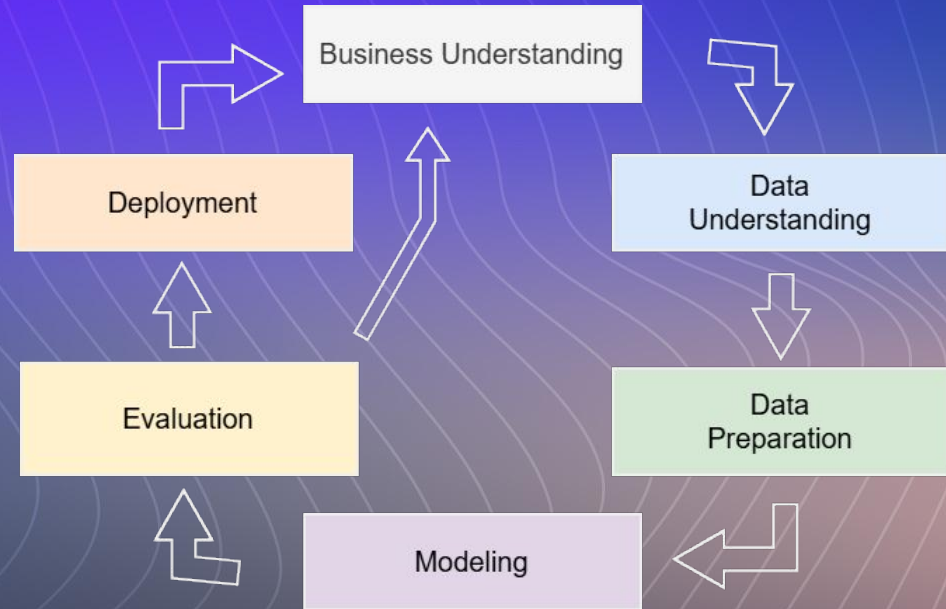
- Berapa target produksi yang realistis?
- Seberapa besar kepuasan pelanggan?
- Apakah operasional organisasi sudah efisien?
- Seberapa besar pangsa pasar yang tersedia?
- Apakah target produksi dapat tercapai?
- Apakah sumber daya sudah sesuai?
- Apakah harga produk sudah sesuai?

Understand business needs

Jenis data yang diperlukan

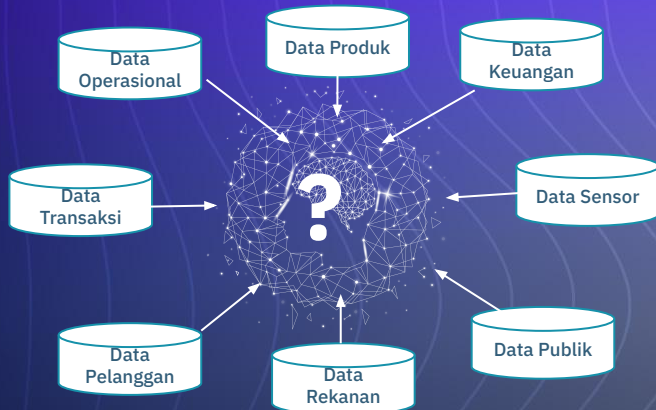
Pertanyaan bisnis yang harus dijawab

Tujuan pengambilan keputusan yg ingin dicapai



Data sources for making decisions

- Menyimpan data dari berbagai sumber
- Mengintegrasikan data dari berbagai sumber
- Menyediakan data yang akurat dan konsisten
- Menyediakan data yang mudah diakses dan digunakan
- Menyediakan data yang aman dan terlindungi



A. DATA ECOSYSTEM

Data Operasional	Data Produk	Data Keuangan	Data Sosial Media
Data Rekanan	Data Pelanggan	Data Transaksi	Crawling, Web Scraping, News

B. DATA EXPLORATION

Identify and combine the data needed to build an analysis model.

Chatbot AI				HR	BI
Create Text	Create Documents	Create Excel Sheet	Create Generate System	Generate	Statistic
1. SOP 2. Scheduling 3. Personal Employee Checking	1. Application Letter 2. Official Travel Letter 3. Contract Documents 4. Plan Documents	1. Offering Letter	1. Create Email Company 2. Automate Invitation 3. Automate Booking	1. Recruitment 2. Verification Scanning	1. Product 2. HR 3. Company Condition

C. PRE-PROCESSING

Fitur Selection, Cleansing, Scaling, Normalization

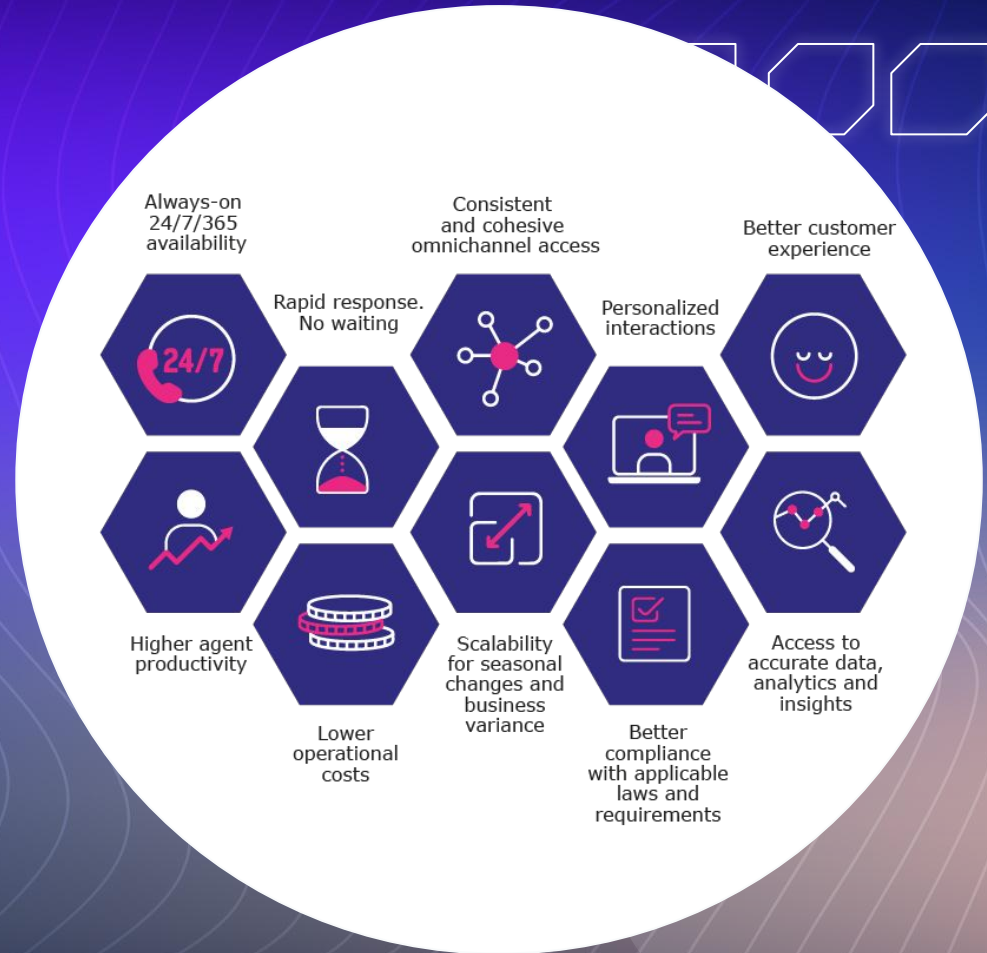
D. PROCESSING



E. EVALUATION

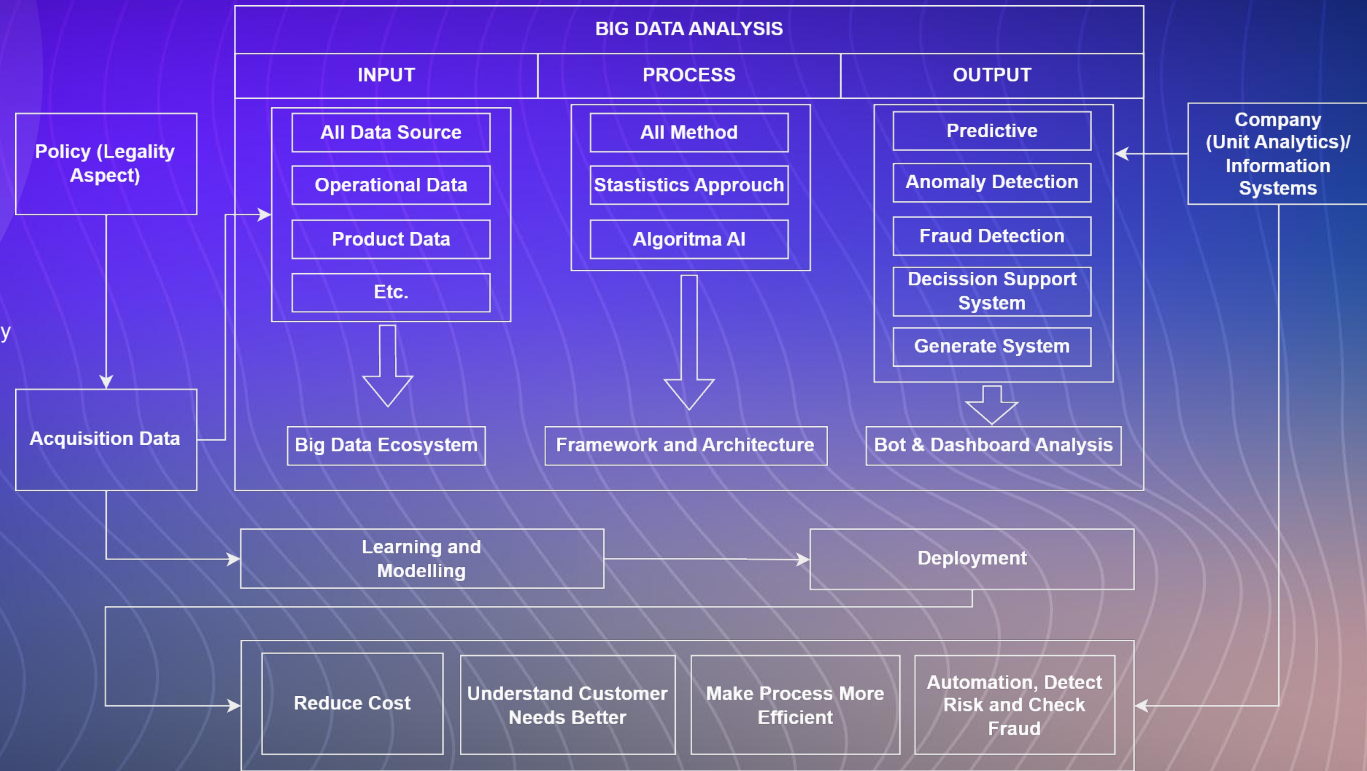
Accuracy, Precission, Recall, MSE, Log Interaction

Challenges And Demands



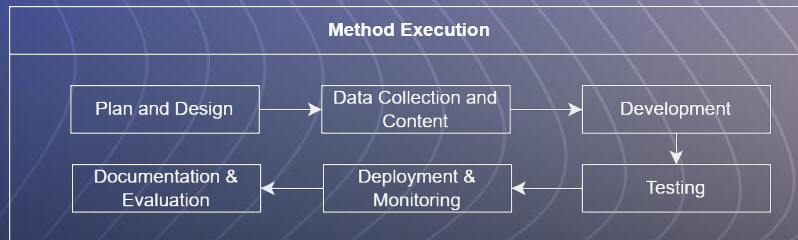
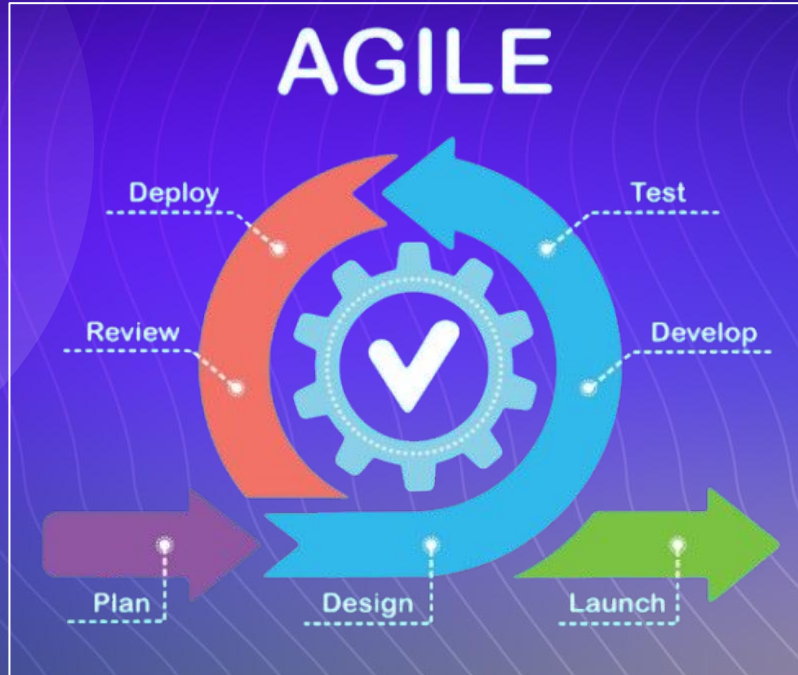
Big Data Development Strategy

1. Building the system with independent domain and supported by Policy Makers. Sustainable development is implemented by the AI Department with supervision from GM.
2. The Learning Process (Model and Method) Analysis) is carried out by the AI Department on an ongoing basis.
3. Collaboration with Experts/Academics
4. Determination of the best, tested and valid analysis model/method.
5. Application of Analysis Models in Big Data Ecosystem.
6. Application of analysis results to internal programs
7. Model Evaluation and Continuous Improvement
- 8.



Method Execution

Agile adoption
And how to execution

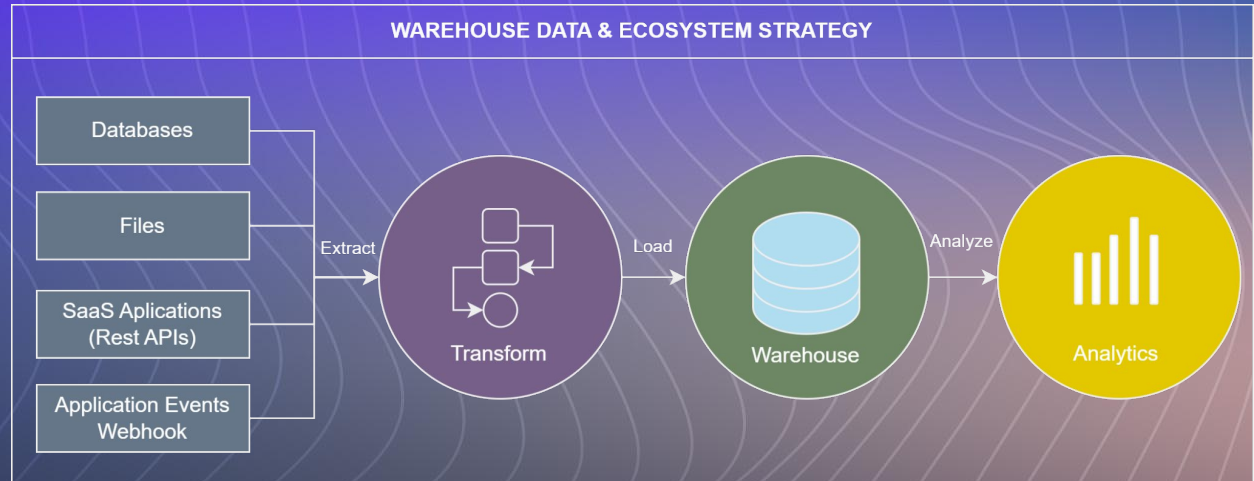
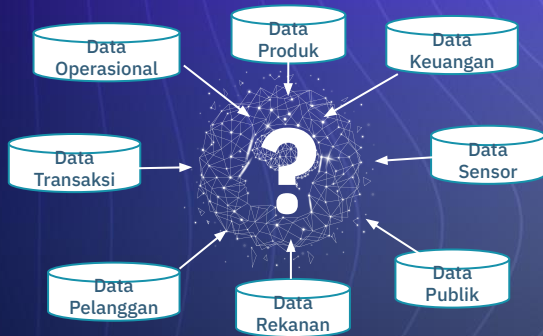


QUESTIONS ABOUT YOUR CHATBOT IDEA

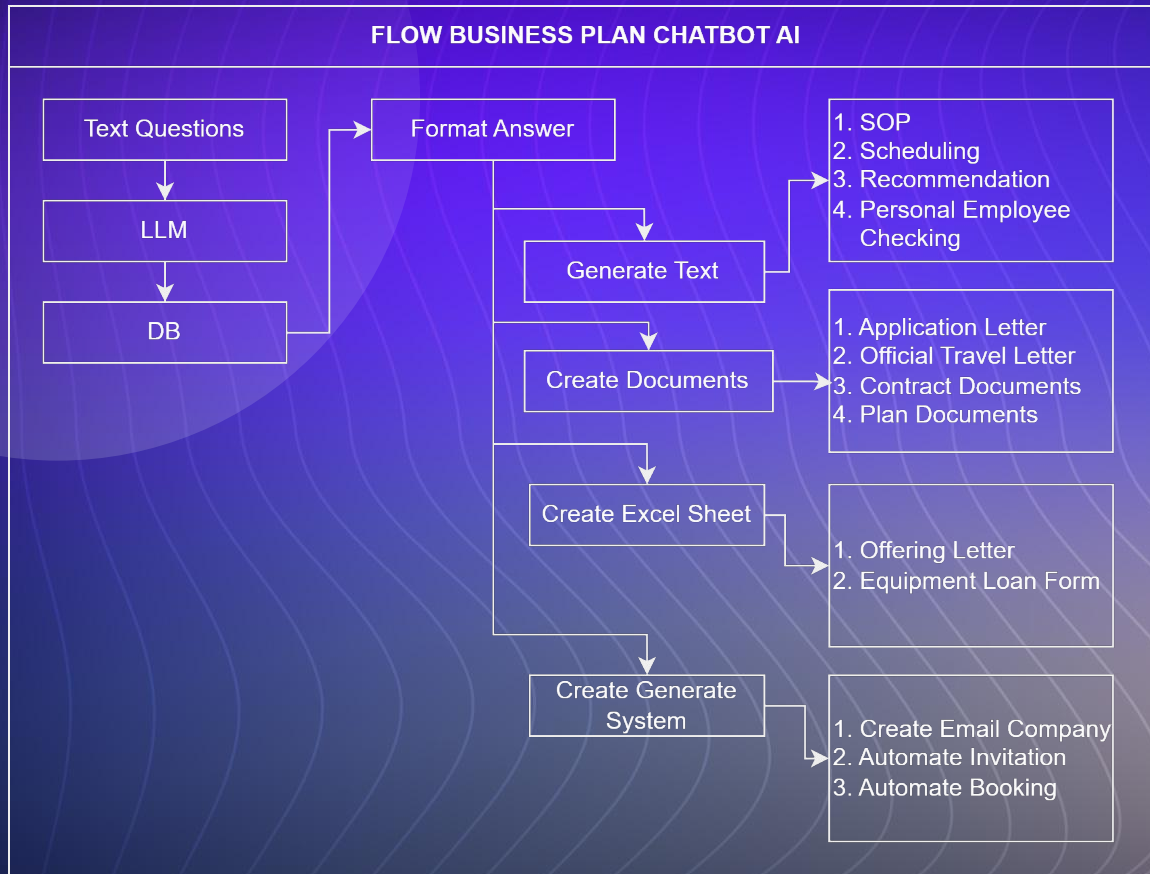
- What is the purpose of your chatbot?
- What are the key goals of your chatbot?
- How will you measure your chatbot's success?
- Who is going to use your chatbot?
- Where will your chatbot be deployed?
- What will your chatbot do?
- Will the chatbot have a character?
- How will the chatbot create value?
- How will people find the chatbot?
- How will you look after your bot?

1. Menentukan sumber data dan cara aksesnya
2. Menentukan transformasi data
 - Cleansing
 - Filtering
 - Agregasi
 - Joining
 - Validation
 - DerivationDll
3. Menentukan cara load

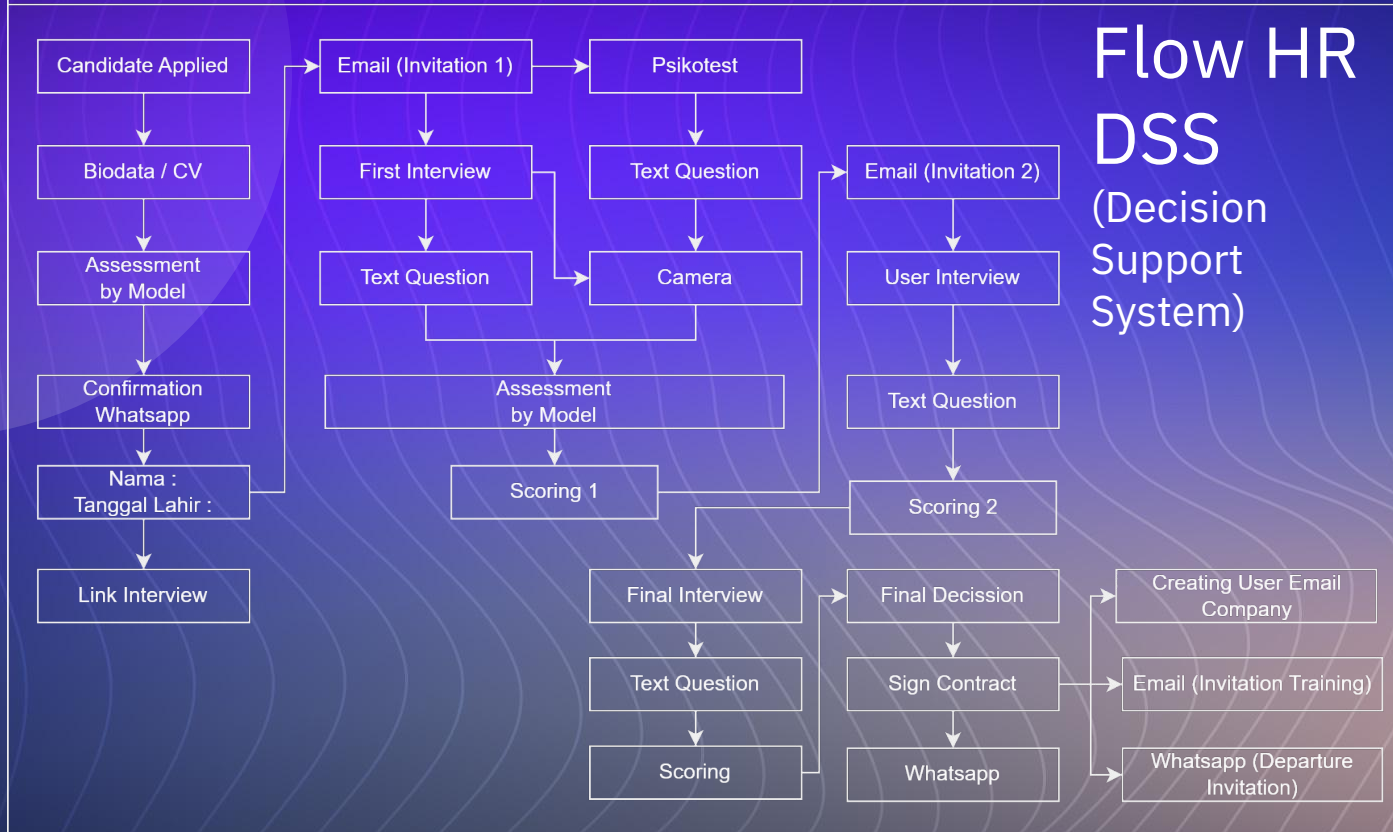
ETL Data & Ecosystem



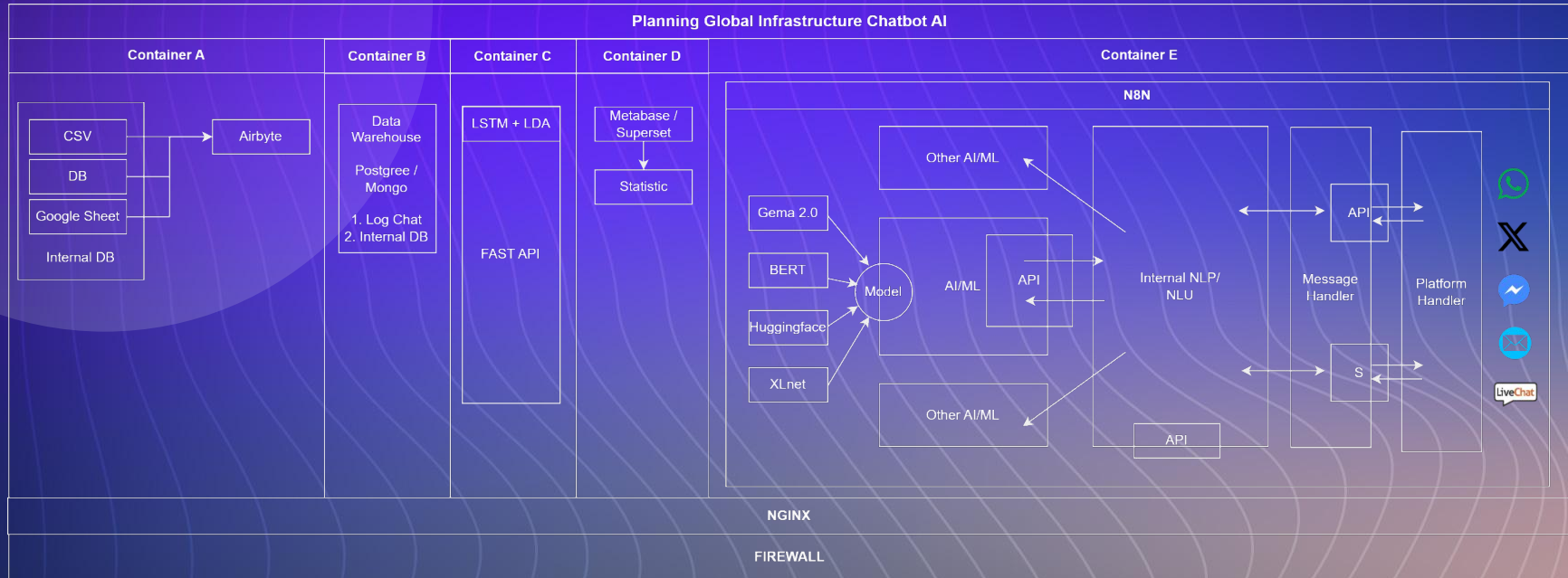
Flow Output Chatbot AI



FLOW BUSINESS HR REQRUITMENT USING AI



Infrastructure Design





Terimakasih